



NOVA

Speak • Command • Automate

NOVA

Intelligent Voice Recognition & Automation Assistant



A Windows-based voice automation assistant that executes system tasks using voice commands, speech recognition, and Windows Registry integration.



Industrial Training Project
Team: NOVAS



Transforming Voice Commands into Intelligent Automation.

❖ Team Members & Roles :

Members	Roles
Bhoomi Sapke	Scrum Master, PO , DEV
Vaishnavi Teli	QA
Unnati Thakur	DEV
Mugdha Lad	DEV
Gayetri Wadile	QA
Krishna Chaudhari	DEV

❖ Project Overview

NOVA (Intelligent Voice Recognition & Automation Assistant) is a Windows-based voice automation system designed to simplify everyday computer tasks through voice commands.

It enables users to:

- Interact with the computer using voice.
- Launch installed applications using the **Windows Registry**.
- Open websites dynamically without hard-coded URLs.
- Perform email automation (send, read, manage).
- Access files and folders.
- Create reminders and timers.
- Receive spoken responses using Text-to-Speech.

❖ Key Features of NOVA

1. Voice Recognition
2. Registry-Based Application Launcher
3. Windows Startup Integration
4. Dynamic Website Opening
5. Email Automation
6. File & Folder Management
7. Reminder & Timer
8. Text-to-Speech
9. Sleep/Wake Mode
10. Language Selection
11. Voice Customization
12. User Preferences
13. Activity Logging

1. Voice Recognition

Purpose

To capture the user's voice through the microphone, convert it into text, and forward the recognized command for further processing

Functional Requirements

- ❑ Capture voice input through the microphone.
- ❑ Convert speech into text accurately.
- ❑ Support continuous voice interaction.
- ❑ Filter background noise where possible.
- ❑ Handle unrecognized speech gracefully.
- ❑ Forward the recognized text to the Command Processing Module.

2. Registry-Based Application Launcher

Purpose

To launch installed Windows applications by retrieving their information from the Windows Registry instead of using hard-coded file paths.

Functional Requirements

- ❑ Search the Windows Registry for installed applications.
- ❑ Retrieve the executable path of the requested application.
- ❑ Launch applications without using hard-coded paths.
- ❑ Display a message if the requested application is not found.
- ❑ Support launching Windows and third-party applications.
- ❑ Ensure compatibility across different Windows systems.

3. Windows Startup Integration

Purpose

To automatically start the NOVA assistant when the user logs into Windows using the Windows Startup Registry.

Functional Requirements

- ❑ Register NOVA in the Windows Startup Registry.
- ❑ Launch the assistant automatically after user login.
- ❑ Retrieve the current Windows username.
- ❑ Display a personalized welcome message.
- ❑ Run the assistant in the background after startup.
- ❑ Allow users to enable or disable auto-start.

4. Dynamic Website Opening

Purpose

To open websites dynamically based on the user's voice command without storing hard-coded website URLs.

Functional Requirements

- ❑ Accept website names through voice commands.
- ❑ Identify the intended website.
- ❑ Open the website in the default web browser.
- ❑ Handle invalid or unavailable websites gracefully.
- ❑ Support commonly used websites.
- ❑ Avoid hard-coded website URLs.

5. Email Automation

Purpose

To enable users to manage emails using voice commands.

Functional Requirements

- ☐ Read unread emails.
- ☐ Compose emails using voice input.
- ☐ Send emails securely.
- ☐ Read email subjects aloud.
- ☐ Search emails based on user requests.
- ☐ Notify users about newly received emails.

6. File & Folder Management

Purpose

To help users access and manage files and folders through voice commands.

Functional Requirements

- ☐ Search files and folders.
- ☐ Open files and folders.
- ☐ Create new folders.
- ☐ Rename files and folders.
- ☐ Delete files with user confirmation.
- ☐ Navigate to requested directories

7. Reminder & Timer

Purpose

To help users manage tasks by setting voice-based reminders, timers, and scheduled actions for applications and system tasks.

Functional Requirements

- ❑ Set reminders and timers using voice commands.
- ❑ Schedule applications or files to open at a specific date and time.
- ❑ Provide voice notifications when reminders or scheduled tasks are due.
- ❑ Allow users to view, edit, or cancel reminders and scheduled tasks.
- ❑ Store reminder and schedule information for future execution.

8. Text-to-Speech Response

Purpose

To provide spoken responses and confirmations for user commands.

Functional Requirements

- ❑ Convert text responses into speech.
- ❑ Speak system notifications.
- ❑ Read assistant responses aloud.
- ❑ Support different voice profiles.
- ❑ Adjust speech rate and volume.
- ❑ Produce clear and natural voice output.

9. Sleep / Wake Mode

Purpose

To control when the assistant listens for voice commands and reduce unnecessary resource usage.

Functional Requirements

- ❑ Enter sleep mode on user command.
- ❑ Resume listening using the wake command.
- ❑ Ignore commands while in sleep mode.
- ❑ Indicate the current listening status.
- ❑ Resume normal operation after activation.
- ❑ Prevent unnecessary microphone processing.

10. Language Selection

Purpose

To allow users to select their preferred language for interacting with the assistant.

Functional Requirements

- ❑ Display available language options.
- ❑ Allow users to change the language.
- ❑ Store the selected language in the NOVA database.
- ❑ Retrieve the selected language from the NOVA database during startup.
- ❑ Apply the selected language to voice responses.
- ❑ Support future language additions.

11. Voice Customization

Purpose

To allow users to personalize the assistant by selecting their preferred voice..

Functional Requirements

- ☐ Provide multiple voice options.
- ☐ Allow users to preview available voices.
- ☐ Store the selected voice in the NOVA database.
- ☐ Retrieve the selected voice from the NOVA database.
- ☐ Apply the selected voice to all responses.
- ☐ Support future voice profiles.

12. Activity Logging

Purpose

To record assistant activities for monitoring, debugging, and future analysis.

Functional Requirements

- ☐ Log executed commands.
- ☐ Record system events.
- ☐ Store error messages.
- ☐ Record application launch history.
- ☐ Maintain activity timestamps.
- ☐ Allow log review for troubleshooting.

13. User Preferences

Purpose

To allow users to personalize the NOVA assistant by saving and automatically applying their preferred settings for a customized experience.

Functional Requirements

- ☐ Save the user's preferred language.
- ☐ Save the selected voice profile.
- ☐ Store the auto-start (startup) preference.
- ☐ Save user-specific assistant settings.
- ☐ Load all saved preferences automatically when NOVA starts.
- ☐ Allow users to update or reset preferences at any time.

❖ Command Processing(Internal Module)

Purpose

Acts as the central controller of the assistant by analyzing recognized voice commands, identifying the user's intent, and routing commands to the appropriate module.

Functional Requirements

- ❑ Receive recognized text from Voice Recognition.
- ❑ Identify the user's intent.
- ❑ Forward the command to the correct module.
- ❑ Handle invalid or unsupported commands.
- ❑ Return the response to Text-to-Speech.

❖ User Interface (GUI) Future Scope

Purpose

To provide a simple interface for displaying the assistant's status and voice activity.

Concept Features

- ❑ Receive recognized text from Voice Recognition.
- ❑ Identify the user's intent.
- ❑ Forward the command to the correct module.
- ❑ Handle invalid or unsupported commands.
- ❑ Return the response to Text-to-Speech.



Status: Monitoring

"Hey Username"

This is a proposed user interface for future implementation. The current version of NOVA is a backend-based application without a graphical user interface (GUI).

❖ Setting Management

Purpose

Allows users to modify assistant settings through voice commands.

Functional Requirements

- ❑ Open the settings menu when the user says "Open Settings" or "Change Settings".
- ❑ Display the settings menu.
- ❑ Change the application Language.
- ❑ Select the Voice (Male/Female).
- ❑ Save updated settings for future sessions.
- ❑ Save all updated settings in the NOVA database.

❖ Developers Task Distribution

DEVELOPERS	TASK 1	TASK 2	TASK 3	TASK 4	ESTIMATED EFFROTS
Bhoomi Sapke	Voice Recognition & Text-To-Speech (4 Hrs)	Command Processing (3 Hrs)	Sleep/Wake (2 Hrs)	Code Review & Final Integration (2 Hrs)	Total: 13 Hrs
Krishna Chaudhari	Windows Registry Integration (4 Hrs)	Windows Startup (3 Hrs)	File & Folder Management (2 Hrs)	Application Launcher (4 Hrs)	Total: 13 Hrs
Unnati Thakur	Website Opening (4 Hrs)	Email Automation (2 Hrs)	Reminder & Task scheduler (3 Hrs)	Activity Logging (4 Hrs)	Total: 13 Hrs
Mugdha Lad	User Preferences (3 Hrs)	Language Selection (4 Hrs)	Voice Customization (3 Hrs)	Setting Management (3 Hrs)	Total: 13 Hrs

❖ QA Task Distribution

QA / TESTER	ASSIGNED DEVELOPERS	TASK 1	TASK 2	WORKING HOURS
Vaishnavi Teli	Bhoomi Sapke, Krishna Chaudhari	Prepare Test Cases (3 Hrs)	Defect Reporting, Retesting (3 Hrs)	Total: 6 Hrs
Gayetri Wadile	Unnati Thakur, Mugdha Lad	Prepare Test Cases (3 Hrs)	Defect Reporting, Retesting (3 Hrs)	Total: 6 Hrs

NOTE : Each QA is responsible for testing the modules developed by their assigned developers and ensuring they meet the project requirements.

❖ NOVA SYSTEM MODULES

Core Modules

- Voice Recognition Module
- Command Processing Module
- Windows Registry Module
- Windows Startup Module
- Application Launcher Module
- Dynamic Website Navigation Module
- Email Automation Module
- File & Folder Management Module
- Reminder & Timer Module

User Interaction Modules

- Text-to-Speech (TTS) Module
- Sleep / Wake Mode Module
- Language Management Module
- Voice Customization Module
- User Preferences Module

System Modules

- Activity Logging Module
- Settings Management Module
- Main Integration Module

❖ Database Design & Development Guidelines

Purpose

- ❑ Define the SQL database structure for NOVA.
- ❑ Ensure consistent database design across all modules.
- ❑ Support secure and efficient data storage.
- ❑ Simplify final project integration.

Developer Guidelines

- ❑ Database Name: **NOVA**
- ❑ Create a local NOVA database.
- ❑ Create only the tables assigned to your module.
- ❑ Follow the predefined table schema.
- ❑ Use the common **database.py** connection module.
- ❑ Test your module with your local database before integration.

1.User_Setting(Setting Management)

Column	Type
setting_id	INT (FK)
language	VARCHAR(30)
voice	VARCHAR(30)
auto_start	BOOLEAN

2.Reminders(Reminder & Task Scheduler)

Column	Type
reminder_id	INT (PK)
title	VARCHAR(100)
reminder_time	DATETIME
status	VARCHAR(20)

3.Activity_Logs(Activity Logging)

Column	Type
log_id	INT (PK)
command	TEXT
status	VARCHAR(20)
timestamp	DATETIME

❖ Future Scope

- ❑ AI Integration using **Google Gemini API**
- ❑ Natural Language Conversation
- ❑ AI-based Question Answering
- ❑ Context-Aware Response Generation
- ❑ AI-powered Code Generation Assistance
- ❑ Proper GUI Integration